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Date: Page:

6. 11(b) Find minimum cost spanning tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

struct Edge{
        int u,v,weight;
};

struct subsets{
        int parent;
};

int find (struct subset subsets[], int i){
        if (subsets[i].parent != i)
            subsets[i].parent = find (subsets, subsets[i].parent);
        return subsets[i].parent;
    }

void unionSet (struct Subset subsets[], int x, int y){
        int rootX = find subsets, x);
        int rootY = find subsets, y);
        subsets[rootX].parent = rootY;
    }

int compare (const void*a, const void*b){
        return ((struct edges *) a)->weight -((struct edges*)b)->
        weight;
    }
```

```
void kruskal (struct Edge edges[], int v, int E){
    struct Edge result [MAX];
    struct Subset subsets[MAX];

    qsort (edges, E, sizeof (edges[0]), compare);

    for (int i=0; i<v ; i++){
        subsets[i].parent = i;
    }
    int e=0;
    int i =0;

    while (e<V-1 && i<E){
        struct Edge next = edges[i++];

        int x = find (subsets, next.u);
        int y = find (subsets, next.v);

        if (x != y){
            result[e++]= next;
            unionSet (subsets, x, y);
        }
    }

    printf("Edges in MST:\n");
    int total =0;
    for (i=0; i<e ; i++){
        printf("%d -- %d == %d\n", result[i].u, result[i].v,
            result[i].weight);
        total += result[i].weight;
    }
```

```c
    printf("Total weight = %d\n", total);
}
int main(){
    int V, E;
    printf("Enter number of vertices & edges: ");
    scanf("%d %d", &V, &E);

    struct Edge edges[MAX];
    printf("Enter edges :\n");
    for(int i=0; i<E; i++){
        scanf("%d%d%d", &edges[i].u,&edges[i].v,&edges[i].weight);
    }
    kruskal(edges, V,E);
    return 0;
}
```

output:

```
Enter number of vertices & edges : 4 5
Enter edges:
0 1 1
1 2 2
0 2 3
2 3 4
1 3 5

Edges in MST :
0--1 ==1
1--2 ==2
2--3==4
Total weight : 7.
```